

Explainable Risk Prediction on Streaming Data: a LSTMs application to SARS-CoV-2 Patients

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Abstract—The evolution of SARS-CoV-2 disease patients differs greatly depending on their demographic and clinical history. The sudden deterioration of some patients requires prompt early detection to reduce mortality. We demonstrate that a Long-Short Term Memory deep learning approach can predict effectively the risk of SARS-CoV-2 patients developing critical illness based on historical clinical data. In this study, we developed this model using a cohort of 2301 patients diagnosed with SARS-CoV-2 using RT-qPCR from February-May 2020. The patients are from several medical centers of Community of Madrid (Hospitales de Madrid del grupo HM Hospitales). The developed model achieves a f1-score of 92.29% and accuracy of 95.94%. In addition, we include model explainability by analyzing robustness to perturbations, feature importance using Shapley's additive explaining technique, and transparency metrics and visualizations to improve the reliability of the AI system and achieve early adoption by clinicians. To provide a clinical tool to the end user we also created an online application for early assessment of patients' risk of serious illness. This proposed rapid response methodology enables efficient allocation of healthcare resources and addresses some pandemic-related challenges.

Index Terms—temporal series, model explainability, risk prediction, rapid crisis response, COVID-19

I. INTRODUCTION

The efficient use of hospitals' and medical centers' resources is critical in COVID-19 pandemic for early treatment and patient recovery. COVID-19 has shown a sudden progression to critical illness in 6.5% of cases and with a mortality rate of 49% [7, 24]. The rapid growth in the number of cases, and specially those that are critical, has posed an important challenge on health systems and its resource management [15] because of its unknown behaviour. The early detection of these patients would improve prognosis ensuring the appropriate care as soon as possible and the use of machine learning models is appealing due to their capacity to learn from data. Overall, this situation is not particular of SARS-CoV-2 disease but can be generalized to any other disease that covers the following criteria: a) Is predictable based on patients' data and b) sensitive to a patient health condition history. These

works also imply a shift from reactive to preventive, and from a global to personalized medicine as stated by the European Union [1].

Different studies have used different techniques to explore, predict, and develop models for risk assessment. In [21] a descriptive observational study of national-wide registry including 109 Spanish hospitals was performed using clustering techniques to explore the dataset and obtain some insights into the four obtained groups. This study did not create any model to predict the risk of a patient but its main contribution is the inclusion of four phenotypic clusters and its analysis to obtain insights into the relation between symptoms and demographic parameters and prognosis. These insights are highly valuable to develop indirect risk prediction models despite their absence in that study. In [26] an importance criteria is used to obtain significant variables ordered by relevancy and then a regression model is built to predict patients' risk. The study used univariate analysis and a set of partial manually implemented methods to obtain a final model based on three previously defined risk level categories. On the other hand, in [12] the authors present a deep learning based survival model to automatically predict the risk of COVID-19 critical illness. In that study the authors predict the survival probability at a given time t (also called *Hazard* function). This approach considers ten statistically significant features (which been previously selected based on significance analysis) to build an hybrid deep neural network and Cox regression model to improve the simplistic Cox regression approach by adding non-linearity among the covariates.

In this study we aim to complement the previous studies to predict survival curves but we considered the temporal dimension (note that none of the previous studies did). Thereof, we use the historical clinical data of a patient to predict the risk illness. To include the timeline, our method fuses historic information (antecedents, procedures, emergencies and so on) along with n-days of evolution data (treatments, tests, and vitals). This information is interpreted with a Long Short Term

Memory (LSTM) recurrent neural network, where an encoding of historic information stands as the memory states and the evolution modifies the given state. It permits handling mid term dependencies (up to 40 days) and the evolving patient's risk health can be extracted by reading the LSTM states. Recall that LSTM has been previously used successfully in medical applications (e.g. in [17] it is used to predict diabetes and mental health from historical patient records).

One common drawback associated to the use of machine learning techniques in medicine is their lack of interpretability or the ability of a model to explain why it has given an output [3]. As presented by the authors in [19] the clinical and end-users are asked to rely on powerful but intrinsically opaque machine learning models. This lack of transparency may imply (1) a lack of quality assurance, (2) failures to elicit trust, and (3) restrictions on physician-patient dialogue. From the perspective of clinicians, a model's output is not very meaningful if it can't be trusted.

In AI context, trust relates to several different aspects that are being study under the XAI (eXplainable Artificial Intelligence) topic. There is no global consensus on which aspects are relevant, but some widely-used characteristics are interpretation of the prediction model's [13], transparency, and robustness [18]. To improve trustworthiness (at least partially) we propose to approach XAI considering three different perspectives. First, a feature importance analysis using SHAP (SHapley Additive exPlanation) technique as suggested at [13] that estimates the importance of each input feature. Second, we test the robustness of the model using a perturbation approach to reassure (at what extend) that a possible change in the inputs will not imply a change in the output decision of the system. And third, we support transparency by providing visual information comparatives of individual patients and its evolution throughout the time.¹

II. DATASET AND METHODOLOGY

The data has been collected from 1944 anonymous patients amidst the SARS-CoV-2 outbreak and includes 54338 daily registries. These registries start at patients entering in the hospital event time and finishes either when they are discharged or deceased (categorized as survival or non-survival). All patients that has an uncertain event (e.g. hospital re-allocation) have been censored. Figure 1 shows the distribution of survival and non-survival by age ranges and can be observed that there is a significant increase of non-survival for older people as proved in other studies [16].

Observing the heterogeneous collected data, two categories must be defined: i) evolutionary data, and ii) historical data. The evolutionary data contains registries recorded since the internment of the patient; whereas historical data contains patient medical records and demographic information.

¹Recall that any software intended by the manufacturer to be used for medical purposes is considered a medical device under the Medical Device Regulation (Regulation (EU) 2017/745).

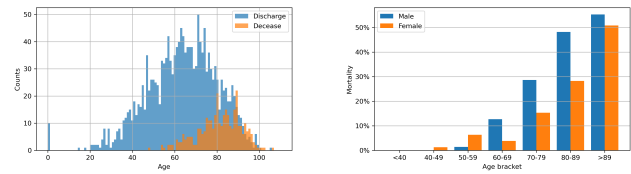


Fig. 1. (left) Frequency histograms for discharged and deceased patients. (right) Observed mortality according to age bracket and biological sex of patients.

A. Evolution data

Every patient is monitored for a variable number of days until an event occurs and the generated medical information is registered. Whenever a patient is sent to the ICU these records are no longer available so ICU periods have been excluded. Evolution information includes vitals (heart rate, temperature, minimal blood pressure, and maximal blood pressure), common medical tests, and treatment dosage. Vitals are aggregated as the maximum, minimum, average, and standard deviation of the daily measurements. 140 patients' tests are also included from an original set of 386 tests. All tests that contain natural language or were conducted less than 10 times were excluded. Moreover some unreadable registries that lack of numerical values were also discarded.

474 daily dosage treatments (number of average dosages) are included using standard medical codes with the first five letters (ATC5 standard) representing the type of medicament and the last two characters the specific treatment (ATC7 standard). Some specific treatments are unknown, but their types are represented, when this happens the end characters are marked as XX. Dosages are set for a time period that is specified in the original dataset. In case of overlapping during a time period, the dosages are added.

B. Historical data

Historical data refers to SARS-CoV-2 patients antecedents and emergencies which are previous to their admission in the hospital. These registries represent the initial state condition of a patient including the suffered emergency to warrant a hospitalization. First, the antecedents details are listed with CIE-10-ES [2] codes. For simplification purposes we use the code before the dot to accurately define the category and nature of the illness allowing its reduction to a set of antecedents of 563. Only the antecedents shared by at least 10 patients are maintained, the others are discarded. Neoplasms and procedures are also filtered with the same criteria allowing a reduction from 72 neoplasms to 3; and from 251 procedures to 43. For records with the same prefix their antecedents are added. All the emergencies are treated with the same preprocessing and from 241 emergency types 34 of them were maintained; there were also 16 procedures, where 5 have been maintained.

C. Methods

We apply a LSTM neural network to the given data including historical registries as the LSTM state and the patient evolution over time is processed throughout the network. Resulting predictions are a measure between 0 and 1, stating the likelihood of decease. This process is detailed in the Annex.

In short, the model, given an arbitrary time window below the maximum stay time (77), will output this likelihood measure. As such we are interested in seeing the accuracy of the model under certain circumstances. First we compute model performance at face value, using all data available at once, trying to determine if a patient survived or not. Next we perform experiments reducing the window with two approaches. The first one begins using a 0 time window (prediction using exclusively historical data) and progressively incrementing the window up to 30 days. The other approach is beginning with a full patient registry and gradually removing days from the beginning. Both approaches aim to see how well the system predicts risk and how well it reacts to missing data (either at the beginning of the time series or at the end).

In this regard we apply Cross-validation (CV) to study the performance of machine learning algorithms to predict the risk using the created SARS-CoV-2 dataset. In standard CV, instances partitioning is based on random sampling of file from a pool wherein all patients are mixed into CV partitions.

We found out trust in such a model is scarce and, as such, we focus on explainability and trustworthiness. We want to visualize the importance of features for each prediction and how each input impacts the predictions. Also, data in hospitals can be faulty, therefore we stress test our system up to 30% randomly missing data. To complement previous explainability efforts we generate visualizations destined to understand the models' behaviour at different time windows.

III. RESULTS

Two thousand two hundred ninety seven patients (medium age 67.89 y.o.; males=1372 medium length of hospital stayed 23.65 days, non-survivors $n = 333$) were analyzed and classified as confirmed when a positive SARS-CoV-2 Real Time-Polymerase chain reaction (RT-PCR) result was obtained. 194 were admitted to intensive care unit (medium length of stay 7.32 days). Demographic and clinical characteristics of patients are shown in Table I. This data was drawn from public HM Hospitals Data Save Lives dataset². The dataset includes demographic, treatments, dosage, vital constants, and laboratory analysis. The classification (survivor or non-survivor) results report is shown in Table II. These results use the patient evolution and clinical history since hospitalization, labelling either if the patient has survived or not. The implemented risk assessment system shows a macro-average f1-score of 92.29% and an accuracy of 95.94%. The minority class (non-survivors) is harder to classify with a 86.99% average f1-score vs. 97.59% for survivors. Standard deviation of f1-scores

²available at via email at covidatasavelives@hnhospitales.com

TABLE I
CHARACTERIZATION OF RELEVANT FEATURES OF THE DATASET.

Characteristics (n=2297)	Overall
Female sex – no (%)	925 (40.25%)
Age – avg [25% quartile, 75% quartile]	67.89 [57, 80]
Stay – avg [25% quartile, 75% quartile]	23.66 [15, 32]
In ICU – no (%)	194 (8.45%)
ICU stay – avg [25% quartile, 75% quartile]	7.32 [1, 10.8]
CIE-10 Coexisting conditions & symptoms – no (%)	
J18 – Pneumonia	874 (38.03%)
J98 – General respiratory symptoms	454 (19.76%)
J96 – Respiratory insufficiency	315 (13.71%)
J12 – Viral pneumonia	194 (8.44%)
B99 – General infectious diseases	118 (5.13%)
B34 – Non-localized viral infection	104 (4.53%)
J22 – Acute lower respiratory tract infection	96 (4.18%)
R05 – Coughing	90 (3.92%)
R50 – Fever	88 (3.83%)
J84 – Other pulmonary diseases	87 (3.79%)
E87 – Other fluid, electrolyte, and acid-base balance disorders	58 (2.52%)
R09 – Other circulatory and respiratory symptoms	48 (2.09%)
R06 – Respiratory alterations	46 (2.00%)
I50 – Cardiac insufficiency	39 (1.70%)
N39 – Other urinary system disorders	32 (1.39%)
N17 – Acute renal failure	32 (1.39%)
I10 – Hypertension	30 (1.31%)
N19 – Renal insufficiency	28 (1.22%)
R19 – Other digestive and abdominal symptoms	27 (1.17%)
Outcome – no (%)	
Survivor	1608 (70.00%)
Non-survivor	333 (14.50%)
Non-discharged	259 (11.28%)
Displaced to hospital	62 (2.70%)
Displaced to social-sanitary center	32 (1.39%)
Voluntary discharge	3 (0.13%)

TABLE II
SUMMARY OF EXPERIMENTAL RESULTS FOR CROSS VALIDATION WITH ALL METRICS MEASURED IN AVERAGED PERCENTAGES WITH THE DEVIATION BETWEEN BRACKETS. SUPPORT FOR EACH CLASS IS THE AVERAGE OF 10-FOLD CV.

	Survivor	Non-survivor	Macro-average	Micro-average
Precision	96.05% (1.5)	95.54% (5.84)	95.79% (3.31)	95.96% (1.9)
Recall	99.19% (1.14)	80.15% (7.88)	89.67% (4.13)	95.94% (1.84)
F1-score	97.59% (1.09)	86.99% (6.12)	92.29% (3.59)	95.78% (1.94)
Support	161	33	194	194
Accuracy	95.94% (1.84)			

across CV runs is low for survivor patients (1.09%) while it is much higher for non-survivors patients (6.12%). On a macro-average level f1-scores have some variability (4.13%), while on accuracy deviation across runs is lower (1.84%).

To get some insights of these overall results, figure 2 shows the macro-average metrics grouped by age ranges and sex. It is worth noting that there is no data for male patients in the 10 to 20 age bracket. To compare if the difference between distributions is relevant, we use the Shapiro-Wilk test to contrast if we can approximate the distribution with a normal distribution. If so, we employ the Welch's t-test and if not, we employ the Mann-Whitney U-Test. We observe that there is not significant differences between male and female patients (accuracy p-value > 0.997; precision p-value > 0.867; recall p-value > 0.940; f1-score p-value > 0.904), having similar degrees of success in the classification. On the other hand it is observed that as age increases, the accuracy decreases (from 100% to 90% mean=93.7/std=5.0 in women; and from 96% to 83% mean=93.25/std=4.9 in men). Results

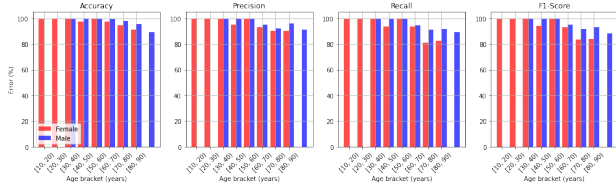


Fig. 2. Breakdown of classification results by age bracket and biological sex of patients. All metrics are included.

are above +95% for all metrics in the range from 10 to 30 years (where there are zero non-survivors) and the accuracy of the model decreases to 82% for males within the range 80 to 90 (the non-survivors percentage in this range is 39.02%).

So far, these results proof classification (survival or non-survival) capabilities of LSTM to detect SARS-CoV-2 given historical records. However, there is a more useful application of the LSTM network model as a risk assessment system during the evolution of the patient. To verify the suitability of the model for this purpose, the evaluation metrics accuracy, f1-score, recall, and precision were evaluated for each day during the hospitalization period (see Fig. 3 left)). As expected, the metrics' series grow as time period passes because the model gathers more information about the patients (SARS-CoV-2 analysis, treatments, or symptoms). For instance, the accuracy metric ranges from 70% at day zero (only the information from the first day hospitalization is used), to over 95% after 30 days of based visibility. This visualization of metrics is useful to show the overall performance of the model as a predictive system, but does not provide insights on the different survival times of individuals. To provide this information Fig. 3 (right), depicts the same set of evaluation metrics but considering the time range from the consequence time event (survived or not survived).

IV. EXPLAINABLE AND AI TRUSTWORTHINESS

This work was proposed as a rapid tool to early response during the pandemic but we experienced a problem of clinic's trust during the early adoption process. Then, we extended the prediction tool to include the explanation of the model with the hope of helping its adoption. Following the definition of explainable AI provided by the authors [4] "Given an audience, an explainable Artificial Intelligence is one that produces details or reasons to make its functioning clear or easy to understand", we developed an AI system that includes: i) feature importance analysis that acts as an insurance that only meaningful features infer the output, i.e. guaranteeing that underlying truthful causality exists in the model reasoning (interpretability), ii) provision of robustness by highlighting potential adversarial perturbations that may occur in the dataset and at what extent they impact in a change of the prediction [4] (robustness), iii) and provision of impartiality in decision-making allowing access to every individual prediction analysis (transparency). All these mean that the interpretation of the system should, in order to be considered practical, provide either an understanding of the model mechanisms and

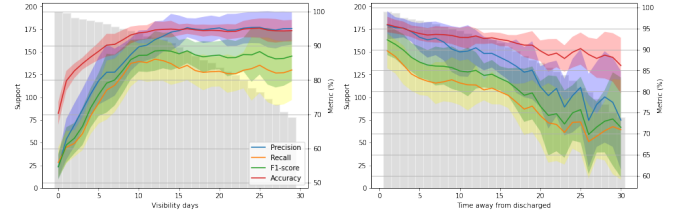


Fig. 3. Evolution of classification metrics given the days seen by the algorithm. Deviation of the 10-fold cross validation is represented by the colored area. The support for each point in the figure is represented by the background bars. (Left) Forward evaluation, aligning patients by number of visible days. (Right) Backwards evaluation, aligning patients by time away from being discharged.

predictions, a visualization of the model's discrimination rules, or hints on what could perturb the model [8].

A. Variable Importance (Interpretability)

SHAP value of feature j on instance x corresponds to the way in which the concrete value of feature j on x modifies the output of the model with respect to other instances. In our case, a DeepLIFT's style of backpropagation is used to obtain SHapley Additive exPlanations (SHAP) values [14]. Note that SHAP is a generalization model for LIME [20], DeepLIFT [23], and layer-wise relevance propagation [22]. SHAP is model agnostic and uses a simple explanation model (e.g. linear models) that approximates locally a sub-part of the complete learned model that is assumed to be linear. Simple models are selected due to they are interpretable by examining their learned coefficients. For a linear model the additive feature attribution method can be expressed by:

$$g(z') = \phi_0 + \sum_{i=1}^M \phi_i z'_i \quad (1)$$

where $z' \in 0,1^M$, M is the number of simplified input features, and $\phi_i \in \mathbb{R}$.

To obtain the overall explanation of the model, we averaged over time the SHAP values of the selected features (please note that daily explanations can also be obtained). Figure 4 shows the most important features according to the generated LSTM model obtained in this study. For antecedents we used directly the 10 most representative and for evolution we obtained the mean absolute value for the evolution features during the whole period (77 days) and for short-term period (1-5 days) of time.

B. Model Resistance to Perturbations (Robustness)

As it is meant for operative environments, it is a concern in medical environments that the use of machine learning models may output unexpected results for new patients. This situation may occur because of missing data that is typical on clinical datasets. Usually, the collecting processes are not under control, specially in a pandemic situation, and we investigate and quantify whether perturbations in the historical records data still satisfy robustness. The natural underlying

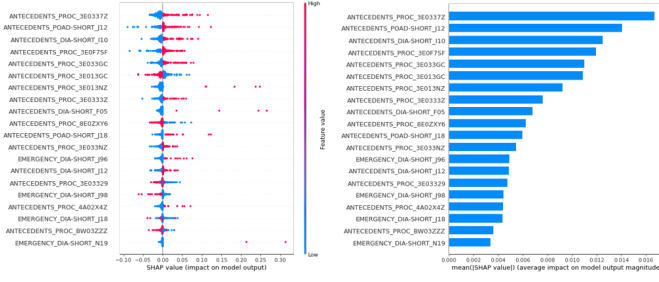


Fig. 4. (Left) Patients' SHAP values for antecedents features. The higher the SHAP value of a feature, the higher is the risk of the deceased. Each point represents a patient and are coloured by the feature value of each instance. The points are accumulated vertically to show the density. (Right) Mean absolute SHAP values for antecedents features (obtained using the best LSTM model output)

question is whether we should trust a model, when it is not robust, and under what circumstances it can be trusted. Hence, there is an expectation that similar input data will lead to similar results, which is a key desideratum for adoption. We employ the definition of (r -robustness) of [5]. For $r \in \mathbb{R}^+$ a model f operating on a point x and outputting a continuous random variable is r -robust to a perturbation δ if and only if

$$\mathbb{P}[\|f(x + \delta) - f(x)\|_2 \leq r] > 0.95 \quad (2)$$

The notion of r -robustness states that a model is robust if, more likely than not, changes in the model's output induced by an input perturbation δ fall within a hypersphere of radius r about the model's outputs on the unperturbed input. First we formalize the intuitive notion of robustness that we seek. We define two scenarios to evaluate robustness: i) how robust is the model to complete days missing data?, and ii) how robust is the model to missing records in the model?

To verify robustness to daily missing data we performed δ perturbations varying from 1 to 10 days and Table III shows the classification percentage relative to the original model as $Z_{r=p}/Z_{r=0}$ where Z is accuracy or F1-Score, and p is in the range of 5% to 30%. Figure 5 also shows the results for accuracy, precision, recall, and F1-score metrics. We establish a threshold of 5% to assure that robustness is maintained.

The second experiment verifies how much missing data is the model able to support whereas maintaining accurate results and robustness. This occurs quite often in medical data due to dropped out patients, unsuccessfully measurement, and/or varied trial design. A widely used approach is to impute missing data with a value (e.g. mean or Bayesian networks [11]) which has been proof to obtain varying results upon the problem and the type of imputation technique [25]. We take advantage of LSTM handling missing data to avoid any of the problems associated to the imputation and possible errors that it may include. Then, we create perturbations in the inputs by removing randomly from the dataset a set of different percentages (5%, 10%, 15%, 20%, 25%, and 30%) values and observed the results to verify the robustness of the model vs. the percentage of missing data. Again we consider a threshold

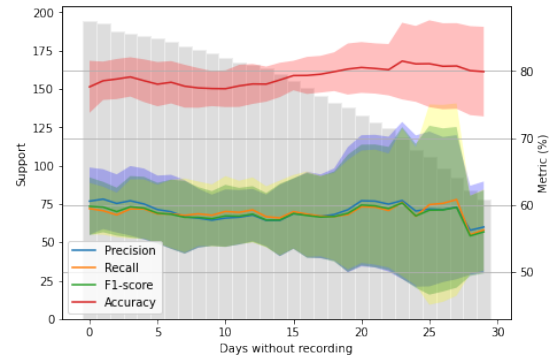


Fig. 5. Results when using a single day of information. The X axis represents how many days have not been recorded for the given patient. Rows with missing data are removed, resulting in less support overall. Results are reliable up to 30 days

of 5% difference in classification to accept that the model is robust. Figure 6 show the results of the different proofs and we can conclude that it is robust up to 15% of missing data. Faults in accuracy start appearing at 10% missing data, as it needs more than 5 days to reach 90% accuracy. With 15% missing data, it is reached at 10 days and further deteriorates to 15 days on 30% missing data. Results never drop below 70% accuracy and, even with faulty information, always improve past 5 days. It is also worth noting that variability (coloured area) across tests increases with the amount of missing data, so the model is more unpredictable as faults are included.

C. Impartiality and completeness in Decision-Making (Transparency)

In [6] authors argue that completeness and interpretability are needed to provide an explicable AI. Moreover there are some detractors [9] of the use of simplified descriptions of complex systems in order to increase trust if the limitations of the simplified descriptions cannot be understood by users, and worse if the explanations are optimized to hide undesirable attributes of the system that may lead to unfounded conclusions. As suggested, for the sake of completeness we decided to complement the validation methodology and take into consideration this point of view of completeness, which describes the performance of a system in an precise way. We included information of patient evolution throughout the days (see Fig.8) and the general representation of this risk assessment for a larger representative number of patients (see Figure 7).

As extracted from the Cambridge Dictionary of English Language, an explanation is "the details or reasons that someone gives to make something clear or easy to understand". In the context of an ML model, this can be rephrased as: "Given a certain audience, explainability refers to the details and reasons a model gives to make its functioning clear or easy to understand." [4]. As introduced, the definition of general objective metrics to assess the quality of XAI approaches is desired and domain specific although it is still an open chal-

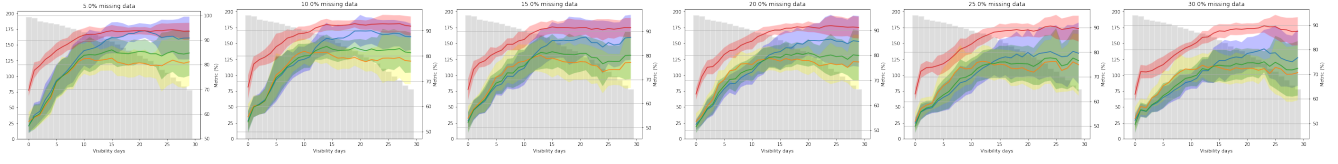


Fig. 6. Results where data is faulty with randomly missing values. Shown in two rows results range from 5% to 30% missing data.

TABLE III
ACCURACY AND F1-SCORE ROBUSTNESS FOR $r=(5, 10, 15, 20, 25, 30)\%$. AVERAGES AND DEVIATIONS ARE REPRESENTED AT THE BOTTOM.

Timestep	r=5%		r=10%		r=15%		r=20%		r=25%		r=30%	
	F1-Score	Accuracy	F1-Score	Accuracy	F1-Score	Accuracy	F1-Score	Accuracy	F1-Score	Accuracy	F1-Score	Accuracy
0	100.86%	98.98%	98.60%	96.35%	95.27%	93.70%	94.00%	91.80%	91.23%	89.68%	90.77%	88.06%
1	96.26%	97.40%	97.94%	96.24%	93.96%	93.11%	89.97%	89.47%	91.46%	90.52%	91.08%	89.09%
2	95.28%	96.26%	96.35%	94.97%	94.58%	92.91%	93.11%	90.97%	91.01%	89.11%	87.33%	85.81%
3	96.62%	96.83%	93.65%	93.79%	92.76%	91.76%	91.11%	89.05%	87.14%	87.11%	86.70%	84.31%
4	93.77%	96.55%	92.92%	93.51%	90.87%	92.24%	90.57%	89.50%	84.71%	85.97%	82.13%	84.01%
5	96.42%	97.03%	94.30%	94.64%	90.49%	91.92%	89.48%	90.88%	82.72%	85.19%	80.96%	83.63%
6	95.01%	96.62%	92.59%	94.32%	87.77%	91.20%	88.89%	90.78%	80.97%	85.24%	79.39%	84.10%
7	96.04%	97.68%	94.79%	96.29%	90.95%	93.63%	88.22%	90.95%	83.52%	87.23%	81.20%	86.02%
8	95.55%	97.86%	93.62%	95.62%	87.75%	92.32%	88.04%	91.59%	83.78%	88.50%	80.37%	86.43%
9	95.42%	97.82%	92.31%	95.30%	87.29%	92.38%	86.05%	91.27%	81.25%	88.12%	79.50%	86.63%
10	96.11%	98.16%	91.76%	94.91%	86.85%	92.39%	85.44%	91.41%	81.47%	88.86%	78.93%	87.07%
11	96.38%	98.25%	91.86%	95.29%	89.74%	94.44%	87.97%	93.10%	84.06%	90.43%	79.89%	88.24%
12	94.85%	97.60%	92.73%	96.22%	90.46%	95.07%	87.96%	93.49%	84.39%	91.13%	81.71%	89.83%
13	95.34%	97.86%	93.69%	96.86%	91.74%	95.96%	89.32%	94.47%	84.90%	91.98%	81.02%	89.90%
14	96.53%	98.45%	94.73%	97.41%	91.48%	96.17%	89.83%	95.01%	85.23%	92.89%	82.91%	91.41%
15	98.12%	99.07%	94.33%	97.34%	93.03%	97.09%	90.22%	95.25%	86.69%	93.41%	84.09%	92.59%
16	95.78%	98.32%	92.85%	96.86%	91.56%	96.39%	90.86%	95.67%	85.88%	93.41%	82.92%	92.13%
17	96.90%	98.79%	94.14%	97.34%	92.23%	96.83%	91.58%	96.05%	86.10%	93.88%	84.00%	93.15%
18	97.34%	99.10%	94.92%	97.75%	92.68%	96.92%	93.20%	96.80%	86.86%	94.37%	85.66%	93.66%
19	97.54%	99.19%	95.71%	98.15%	92.21%	96.69%	92.26%	96.49%	86.52%	93.92%	84.87%	93.40%
20	97.86%	99.30%	94.66%	97.79%	92.67%	96.97%	92.81%	96.86%	86.21%	94.27%	85.55%	93.75%
21	98.50%	99.60%	94.86%	97.95%	93.54%	97.35%	92.92%	96.82%	85.79%	94.06%	86.37%	94.09%
22	97.58%	99.23%	95.64%	98.22%	91.56%	96.48%	91.55%	96.50%	87.30%	94.54%	85.88%	94.31%
23	96.55%	98.91%	93.85%	97.62%	90.30%	96.18%	91.21%	96.66%	88.57%	95.13%	85.86%	94.53%
24	95.60%	98.60%	94.17%	97.53%	87.63%	95.36%	91.75%	96.65%	88.92%	95.31%	83.86%	93.89%
25	95.91%	98.83%	93.39%	97.33%	87.88%	95.84%	90.07%	96.32%	87.59%	94.84%	84.81%	93.86%
26	98.33%	99.56%	94.85%	97.98%	89.19%	96.13%	91.68%	96.74%	85.61%	94.36%	84.67%	93.27%
27	98.14%	99.50%	96.19%	98.31%	90.01%	96.68%	90.96%	96.69%	87.11%	94.87%	83.35%	93.31%
28	97.65%	99.14%	94.47%	97.60%	92.61%	97.15%	92.88%	96.83%	89.90%	95.38%	83.15%	92.58%
29	97.42%	99.08%	93.43%	97.20%	92.06%	96.97%	91.91%	96.54%	87.98%	94.83%	83.24%	92.66%
Avg	96.66%	98.32%	94.31%	96.56%	91.04%	94.94%	90.53%	94.02%	86.16%	91.62%	83.74%	90.19%
Std	1.36%	0.94%	1.54%	1.35%	2.18%	1.96%	2.05%	2.69%	2.69%	3.22%	2.94%	3.59%

lenge [4]. Here we provide a metric and visual comparatives to gain trustworthiness on the LSTM-DNN model:

- The first is based on incremental risk prediction assessment. The number of observed deceased $O(y^+)$ should be correlated to the expected $E(y^+)$ according to the risk values obtained by the model $E(y^+) = LSTM(x)$ for the different days t . That is, as the predicted risk increases, the observed percentage of deceased should also increase being its distribution similar. To measure this similarity in the distributions we split the prediction range values in 10 bins equally sized (0-0.1, 0.1-0.2,...,0.9-1) and measure the correlation between the mean predictive value for all the patients that are within each range ($S1$) and the observed percentage of deceased patients ($S2$). The coherent expected behavior is that as the risk predicted by the model increases, also increases the percentage of patients that deceased. The correlation obtained between both series is $r_{(S1,S2)} = 0.99$ and the p-value under the null hypothesis that both series

have the same correlation mean and distribution is $p\text{-value}_{(S1,S2)} = 1.6E - 7$ rejecting the null hypothesis and concluding that the correlation is significant. This means that the stated hypothesis that "as the risk of the models increases so does the observed deceased" is true.

- The visual comparative allows to better understand patient behavior and explain its evolution. The first visualization explains individual patient risk prediction behavior throughout time (see Figure 8). This visualization allows to compare between different individuals risk predictions. The second (see Figure 7) shows the average risk assessed up to 30 days. This visualization reinforces confidence by indicating that uncertainty is greater during the first 10 days but later the risk boundary is clear between individuals who survive and those who do not.

V. DISCUSSION AND SOME CONCLUSIONS

This paper is an attempt to provide some insights into the medical operating environment, specifically in the case of

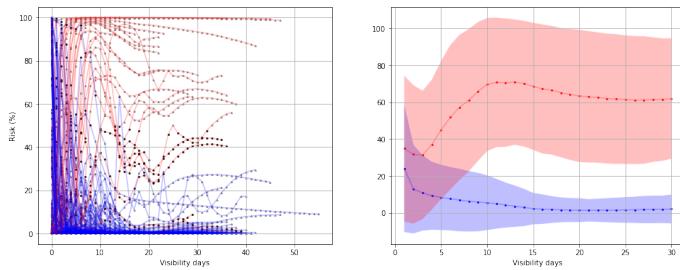


Fig. 7. Left: Random sample of 200 patients (for clarity). Series describe the risk assessed by our network, red and blue lines represent deceased and discharged patients respectively. Grey triangles represent correct assessments, while black dots are for incorrect assessments. Right: Average risk assessed up to 30 days, red and blue are for deceased and discharged patients respectively. The shadowed areas represent the standard deviation. On average, the risk assessment starts at a low value (approx. 30%) for non-survival patients steadily growing for 10 days until reaches a maximum plateau. This means that, after 10 days, adding new information about the patient doesn't increase the risk prediction capabilities of the system. Similarly, survival patients reach the minimum plateau after 15 days. Their standard deviation also overlaps during the first days indicating a higher classification uncertainty (the probability that a non-survival patient may have a lower risk score than other that survive is higher). This effect decreases as the number of historical data increases.

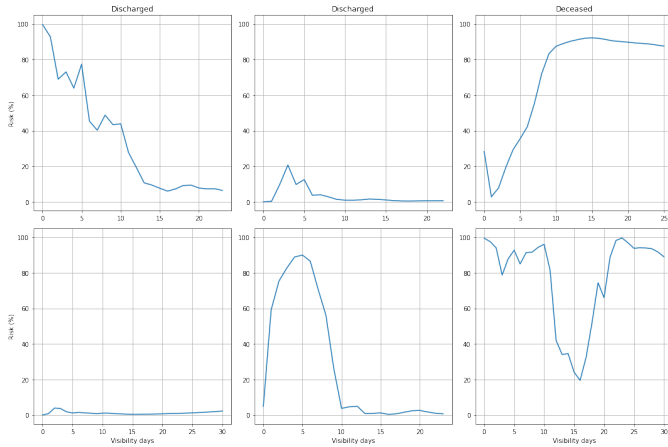


Fig. 8. Risk patient evolution of six representative patients during their whole stay in the hospital (selected to show different typical behaviors). The two first columns represent survival patients and the last column non-survival.

pandemics, and we highlight and propose some problems and solutions related to implementation and explainability in this context. Evaluating patients by age, we found a fair assessment of patient risk. Generalizing, there are not excessive differences when comparing sex. Elderly patients are more difficult to assess than younger patients. This may be due to the higher reliability of the model in discharged patients, together with the lower mortality rate of younger patients. These data also point to differences in the way the model recognizes the age and sex of the patients analyzed. These results are slightly more reliable in younger patients.

The results become more reliable as more data becomes available to the model, meaning that any measure of risk given by the system is better informed than the last assessment made. Accuracy is high from 15 days onwards, and results are fair

in the early stages of screening, with an accuracy of around 75%. Furthermore, if patients are at least a month away from discharge, it can be assumed that the model assesses their risk with an accuracy of better than 80%. As more information is available to the network, the assessment of the patient's status for the current day better approximates the final status (either discharged (< 0.5) or deceased (≥ 0.5)). It should also be noted that the standard deviation of the metrics increases over time, so it is possible that the quality of the model also depends on the amount of data available at different points in time.

The XAI qualitative visualizations show when the model detects low-risk and high-risk cases. Occasionally, the model detects indications of high risk in patients who ultimately survive, which may constitute a medical emergency. The model produces consistent predictions that demonstrate that the risk measurements are reliable. It is also worth noting that the model is very robust against faulty data, withstand 5% missing data without problems, and withstand large amounts of noise (acceptable results up to 30% missing data). In addition, data can be recorded up to 10 days without major loss of reliability. Model-agnostic explainability tools, such as SHAP, are very important in this context for building trust, but there is still a need to automate the process and evaluate their usefulness to clinical and how these tools can improve trustworthiness.

To summarize, results show that the model is reliable to approximate the risk of a patient using the data as presented. As the patient evolves over time the network updates risks based on the treatment and vitals of a patient. This conclusions are examined for a month, where results prove to be highly reliable. After 30 days results begin to deteriorate and no guarantees about reliability can be given as there are not enough patients (alive or deceased) past the 30 day mark to accurately train and evaluate a machine learning model. As next steps, a methodology for validation of the explanations still needs to be addressed to build confidence and support early adoption of such artificial intelligent systems in healthcare.

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REFERENCES

- [1] Notices from european union institutions, bodies, offices and agencies: Council conclusions on personalised medicine for patients (2015/c 421/03). *OJ*, pages 2–5, 2015-12-17.
- [2] eCIE-Maps - CIE-10-ES Diagnósticos, Dec 2019. [Online; accessed 23. Sep. 2020].
- [3] Muhammad Aurangzeb Ahmad, Carly Eckert, and Ankur Teredesai. Interpretable machine learning in healthcare. In *Proceedings of the 2018 ACM International Conference on Bioinformatics, Computational Biology, and*

- Health Informatics*, BCB '18, page 559–560, New York, NY, USA, 2018. Association for Computing Machinery.
- [4] Alejandro Barredo et al. Arrieta. Explainable Artificial Intelligence (XAI): Concepts, taxonomies, opportunities and challenges toward responsible AI. *Information Fusion*, 58:82–115, 2020.
 - [5] Alexander Camuto, Matthew Willetts, Stephen Roberts, Chris Holme, and Tom Rainforth. Towards a theoretical understanding of the robustness of variational autoencoders. *arXiv preprint arXiv:2007.07365*, 2020., 2020.
 - [6] et al. Gilpin. Explaining explanations: An overview of interpretability of machine learning. In *2018 IEEE 5th International Conference on Data Science and Advanced Analytics (DSAA)*, pages 80–89, 2018.
 - [7] Wei-jie Guan, Zheng-yi Ni, Yu Hu, Wen-hua Liang, Chun-quan Ou, Jian-xing He, Lei Liu, Hong Shan, Chun-liang Lei, David SC Hui, et al. Clinical characteristics of coronavirus disease 2019 in china. *New England journal of medicine*, 382(18):1708–1720, 2020.
 - [8] Patrick Hall. On the art and science of machine learning explanations. *arXiv preprint arXiv:1810.02909*, 2018.
 - [9] B. Herman. The promise and peril of human evaluation for model interpretability. *arXiv:1711.07414*, 2017.
 - [10] Sepp Hochreiter and Jürgen Schmidhuber. Long Short-Term Memory. *Neural Comput.*, 9(8):1735–1780, Nov 1997.
 - [11] Estevam Hruschka, Eduardo Hruschka, and Nelson Ebecken. Bayesian networks for imputation in classification problems. *Journal of Intelligent Information Systems*, 29:231–252, 12 2007.
 - [12] W. Liang, J. Yao, Chen A., and et al. Early triage of critically ill covid-19 patients using deep learning. *Nat. Commun.*, 11(3543), Jul. 2020.
 - [13] Scott M Lundberg and Su-In Lee. A unified approach to interpreting model predictions. In I. Guyon, U. V. Luxburg, S. Bengio, H. Wallach, R. Fergus, S. Vishwanathan, and R. Garnett, editors, *Advances in Neural Information Processing Systems*, volume 30. Curran Associates, Inc., 2017.
 - [14] Scott M. Lundberg and Su-In Lee. A unified approach to interpreting model predictions. In *Proceedings of the 31st International Conference on Neural Information Processing Systems*, NIPS'17, page 4768–4777, Red Hook, NY, USA, 2017. Curran Associates Inc.
 - [15] World Helath Organization. COVID-19 Monitoring. <https://covid19.who.int/region/euro/country/es>, 2021. [Online; accessed 14-February-2021].
 - [16] Cardinal-Fernández P, García-Cuesta E, and et al. Barberán J. Clinical characteristics and outcomes of 1,331 patients with covid-19: Hm spanish cohort. *Rev Esp Quimioter.*, 34(4):342–352, 2021.
 - [17] Trang Pham, Truyen Tran, Dinh Phung, and Svetha Venkatesh. Predicting healthcare trajectories from medical records: A deep learning approach. *Journal of Biomedical Informatics*, 69:218 – 229, 2017.
 - [18] A. Preece, D. Harborne, D. Braines, R. Tomsett, and S. Chakraborty. Stakeholders in explainable ai. *arXiv preprint arXiv:1810.02909*, 2018.
 - [19] Thomas P. Quinn, Stephan Jacobs, Manisha Senadeera, Vuong Le, and Simon Coghlan. The three ghosts of medical ai: Can the black-box present deliver? *Artificial Intelligence in Medicine*, pages 102–110, 2021.
 - [20] Marco Tulio Ribeiro, Sameer Singh, and Carlos Guestrin. "why should i trust you?": Explaining the predictions of any classifier. *KDD '16*, page 1135–1144, New York, NY, USA, 2016. Association for Computing Machinery.
 - [21] et al. Rubio-Rivas. Predicting clinical outcome with phenotypic clusters in covid-19 pneumonia: An analysis of 12,066 hospitalized patients from the spanish registry semi-covid-19. *Journal of Clinical Medicine*, 9(11), 2020.
 - [22] Bach S, Binder A, Montavon G, Klauschen F, Müller K-R, and Samek W. On pixel-wise explanations for non-linear classifier decisions by layer-wise relevance propagation. *PLoS ONE* 10(7): e0130140., 2020.
 - [23] Avanti Shrikumar, Peyton Greenside, and Anshul Kundaje. Learning important features through propagating activation differences. In Doina Precup and Yee Whye Teh, editors, *Proceedings of the 34th International Conference on Machine Learning*, volume 70 of *Proceedings of Machine Learning Research*, pages 3145–3153. PMLR, 06–11 Aug 2017.
 - [24] Zunyou Wu and Jennifer M. McGoogan. Characteristics of and Important Lessons From the Coronavirus Disease 2019 (COVID-19) Outbreak in China: Summary of a Report of 72314 Cases From the Chinese Center for Disease Control and Prevention. *JAMA*, 323(13):1239–1242, 04 2020.
 - [25] Bing Zhu, Changzheng He, and Panos Liatsis. A robust missing value imputation method for noisy data. *Appl. Intell.*, 36:61–74, 01 2012.
 - [26] Álvarez-Mon M et al. A Predictive Model and Risk Factors for Case Fatality of COVID-19. *J Pers Med.*, 11(1), 1 2021.

ANNEX: MACHINE LEARNING DEVELOPMENT

A LSTM network has been proposed to solve this problem, interpreting a patient state from their historical data and updating said state through the information gathered during their stay as the hospital. This recurrent neural network is able to capture temporal patterns in time series, being very robust on short (< 100) time windows.

Our network has two different data feeds, as described in subsections II-A and II-B. In our data model, historical data (antecedents, illnesses present on admission, and so on) represents the starting health state of the patient, while the evolution data (treatments, tests and vitals) represents how the initial health state of the patient progresses over time. An adequate way to map this relationship between evolution and historical data, is to use an LSTM [10], which alters states over time to produce outputs.

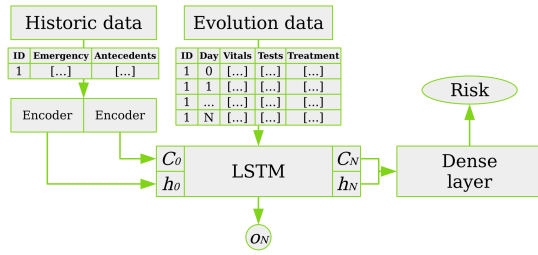


Fig. 9. Visual representation of the pipeline followed, from the historic and evolution data to the risk assessment.

The architecture proposed in figure 9 represents the design of an artificial neural network with a LSTM as the central component. First, given the historical data represents this information explicitly, it is encoded to represent the starting states of the LSTM module. This encoding is done via a dense layer, mapping the initial input shape into an adequate length vector for the LSTM memory states.

The original LSTM formulae that defines the interactions between weights, states, inputs and outputs is represented in Equation 3. Each LSTM neuron has these elements, where x is a vector of inputs, h_t and C_t are the states (short-term and long-term respectively represented via a digit) at a time-step t , o_t is the output at time-step t . Each gate (i_t , f_t , o_t and $\tilde{C}_t$) has two sets of weights named U and W for the input gate (U^i and W^i), forget gate (U^f and W^f), output gate (U^o and W^o) and candidate gate (U^g and W^g). These weights are the parameters trained by the network.

$$\begin{aligned}
 i_t &= \sigma(x_t U^i + h_{t-1} W^i) \\
 f_t &= \sigma(x_t U^f + h_{t-1} W^f) \\
 o_t &= \sigma(x_t U^o + h_{t-1} W^o) \\
 \tilde{C}_t &= \tanh(x_t U^g + h_{t-1} W^g) \\
 C_t &= \sigma(f_t * \tilde{C}_{t-1} + i_t * \tilde{C}_{t-1}) \\
 h_t &= \tanh(C_t) * o_t
 \end{aligned} \tag{3}$$

If an individual example of the historical data is defined as a vector H , the definition of $C_{t=0}$ and $h_{t=0}$ can be found at equation 4. These weights W^C and W^h define the initial state of the any LSTM given a vector of inputs.

$$\begin{aligned}
 C_{t=0} &= \text{relu}(H W^C + \theta^C) \\
 h_{t=0} &= \text{relu}(H W^h + \theta^h)
 \end{aligned} \tag{4}$$

In our design the inputs to a LSTM cell are the registries of patient evolution data. We can define these as x_t as per the LSTM equation, where x_0 is the evolution of the patient at day 0, x_1 is the evolution at day 1 and so on, up to N days. However, in this design, we are not interested in the outputs of the LSTM (o_N), instead our focus is to retrieve the end internal state of the LSTM neurons and operate over them to extract the risk of any given patient. Given that we have encoded the initial state of the patient with the previous health

status of the patient, the states C_N and h_N are transformed by a series of N inputs, in return representing the current state of the patient.

These states can be concatenated and processed through another dense layer to finally produce a binary output (0:[No risk or guaranteed discharge] or 1:[Maximum risk or guaranteed death]). This is shown in equation 5, where the concatenation of all LSTM neuron states h and C are represented as S . In this case K is a vector of processed values and y is the binary output of the network.

$$\begin{aligned}
 K &= \text{relu}(S W^o + \theta^o) \\
 y &= \sigma(K W^y + \theta^y)
 \end{aligned} \tag{5}$$

This composite network will approach to 1 or 0 for high or low risk respectively. Our data has no uncertainty in-between days, meaning that the discharge condition is only known when it actually happens. As per the design of the network, the risk of someone who has already been discharged is 0, while the risk of someone who has already died is 1; therefore someone who has not yet been discharged or who has not died, will have a risk assessment between 1 or 0. When an incomplete set of days is fed to the network, the system is able to recognize the state of the patient and assign a risk factor. This is possible because the design of the network allows to analyse the health state of the patient (both: encoding of initial health state as the state of the LSTM and analysing the end state of the network with a dense layer).

Training

Training the network with unbalanced data as shown in subsection II is not a trivial task. Hyper-parameters such as network size, learning rate and epoch number have been tuned empirically while using techniques for imbalanced data. The network parameters are optimized via stochastic gradient descent (SGD) with a learning rate of 10^{-3} without momentum. The loss chosen to back-propagate has been binary cross-entropy. Weights have been applied to the loss function to recognize the minority class (Maximum risk), with a 0.75/0.25 value for 1/0 respectively. Further regularization of the network has been required: an internal L2 regularization of the LSTM cells' recurrent weights and a batch normalization step just before the output neuron. To ensure the fidelity of the results a stratified 10-fold cross validation has been performed, separating data by patient identification. All results shown in the experimental results section have been extracted via this method. From the CV we extract four metrics at macro-level (weighted by the number of instances in each class), namely Accuracy, Precision, Recall and F1-score.